

Integration

Use APEX. Section 5.1-5.3 today.

Next week. 5.4 and some parts of chap 6.

Won't be doing 5.5. (Numerical Methods module next year).

For us, integration will be introduced as two different concepts, initially unrelated, but then with Fundamental Theorem of Calculus these two will be unified.

① Integration as anti-differentiation, i.e. an inverse operation to the derivative.

eg. $f(x) = x^2 \xrightarrow{\text{diff.}} f'(x) = 2x.$

← anti-differentiation /
(indefinite) integration

② Integration as the process of "summing up" the total accumulated value of a function across some

interval. This will give the "area beneath the curve" or the definite integral of a function.

① and ② will be unified by the F.T.C. Fundamental Theorem of Calculus.

① The anti-derivative

Definition An anti-derivative of a function $f(x)$ is any function $F(x)$ that satisfies.

$$\frac{d}{dx} F(x) = f(x) \quad \text{or} \quad F'(x) = f(x).$$

We can use the notation

$\int f(x) dx$ to denote such a function $F(x)$, called the indefinite integral of $f(x)$.

eg. $F(x) = x^2$ is an anti-derivative or indefinite integral of the function

$$f(x) = 2x.$$

But so is the function $G(x) = x^2 + 5$

since $G'(x) = 2x$

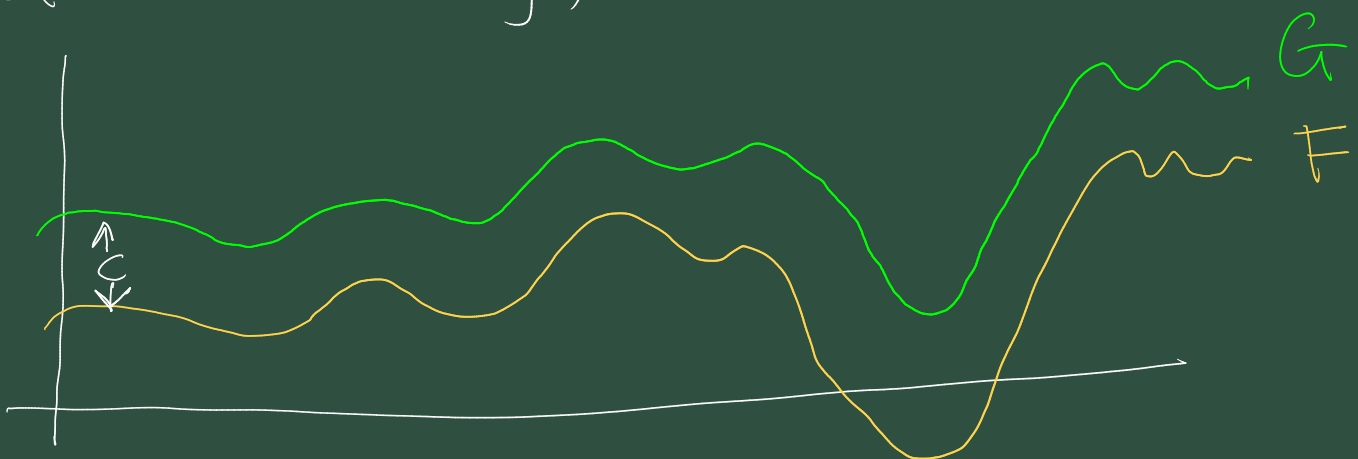
Indeed so is any function $H(x) = x^2 + C$, where C is a constant.

Theorem 5.1.4 If F, G are two indefinite integrals of a function $f(x)$, then.

$$\forall x \in I \quad G(x) = F(x) + C$$

for some constant $C \in \mathbb{R}$.

Geometrically, what this means is.



Using this concept of anti-differentiation
Theorem 5.1.8 gathers many
standard indefinite integrals and
standard properties of integration process.

Other important properties of differentiation
will have useful implications for the
indefinite integral examined in chapter 6
of APEX.

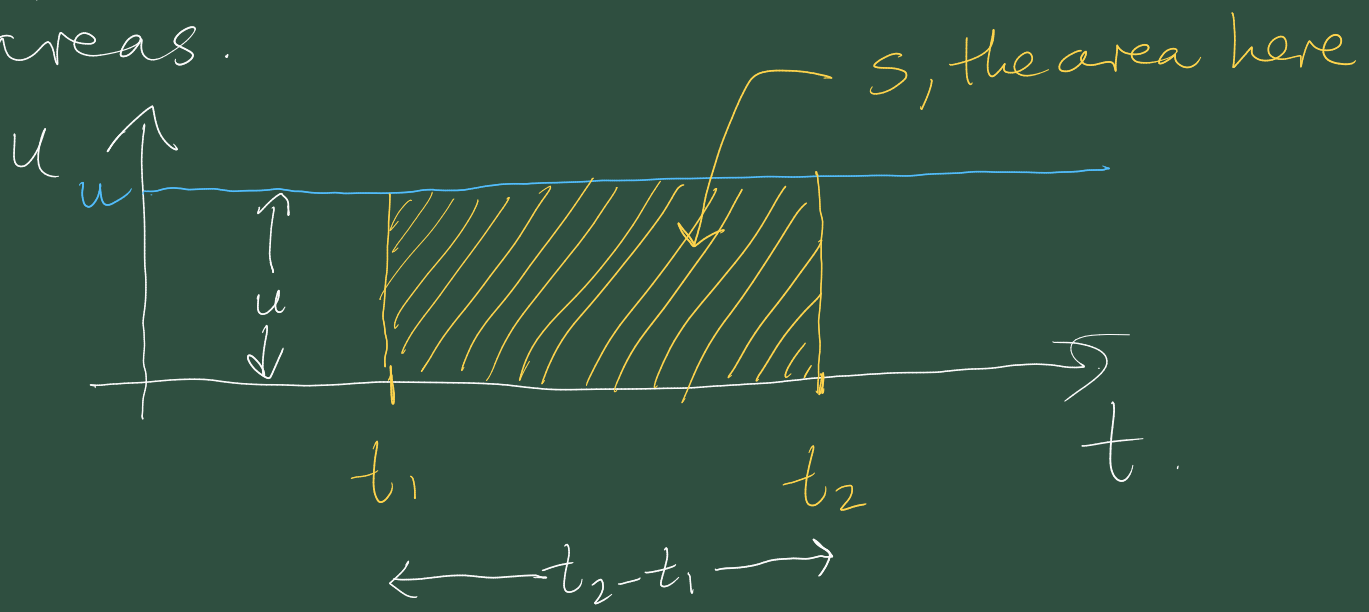
② The definite integral.

Illustrating example (1-dimensional motion)

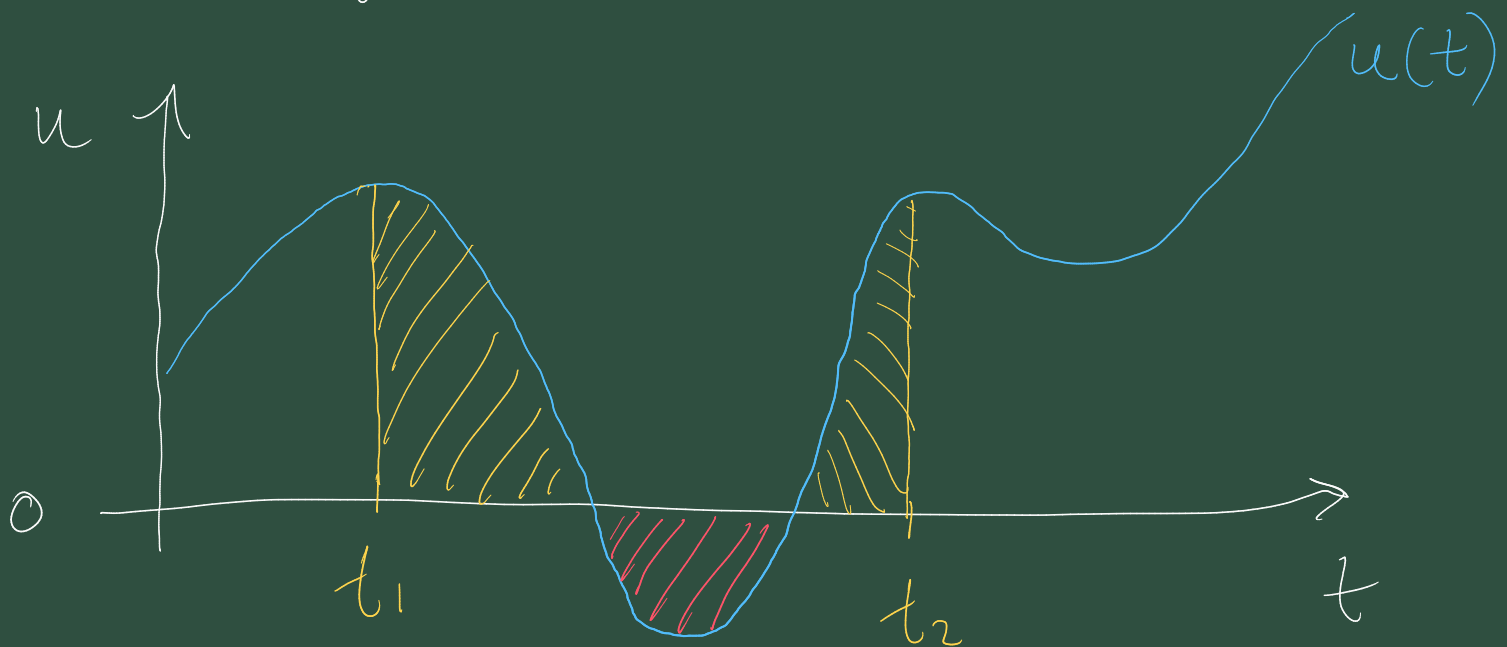
Consider an object travelling with
constant velocity u , beginning at time t_1
ending at time t_2 . Then the displacement
(total distance travelled) s of the object
will be given by

$$s = u(t_2 - t_1)$$

Key idea: products like this can often be conceived geometrically as areas.



This concept extends to situations with changing velocity



The displacement will be the signed area between $u(t)$ and the t -axis with areas above the line counted positively ($u > 0$) and areas below

the line counted negatively ($u < 0$).

In general, this signed area, is known as the definite integral written as

$$\int_{t_1}^{t_2} u(t) dt$$

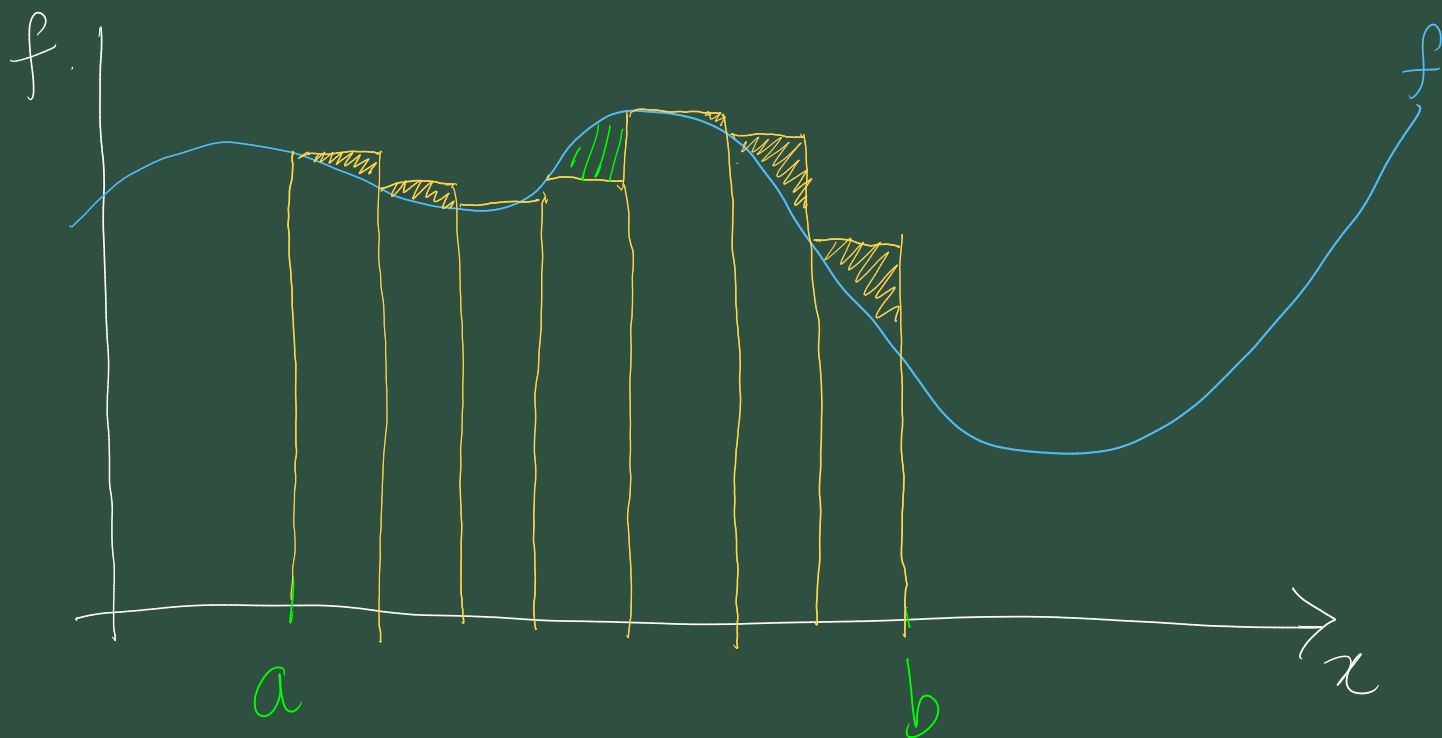
General problem:

Given a variable function $u(t)$
How do we compute $\int_{t_1}^{t_2} u(t) dt$?

For certain nice functions $u(t)$
we can use known geometry of
rectangles, triangles, circles, ...
to evaluate the integrals

↳ Riemann sums / Riemann
integration

Key idea



$$\int_a^b f(x) dx$$

Idea is to approximate the area as the sum of the areas of a collection of rectangles across interval (a, b) , whose heights "match" the height of the function f over the width of the rectangle.

Clearly there is an associated error here. But if we take the limit of such a rectangular sum as

the number of rectangles $\rightarrow \infty$ and
their widths $\rightarrow 0$ then in the
limit we will exactly determine

$$\int_a^b f(x) dx.$$

Continue at 12:00.

Set all this up carefully.

Use a partition/sub-division of $[a, b]$

• $x_0 = a, x_1, x_2, x_3, \dots, x_{n-1}, x_n = b$

where $x_i < x_{i+1}$

• write $\Delta x_{i-1} = x_i - x_{i-1} = \text{width of the } i^{\text{th}} \text{ rectangle}$

for each $i = 1, \dots, n$

• let x_i^* be a representative point

in i^{th} sub-interval $x_{i-1} \leq x_i^* \leq x_i$

• $\Delta x := \max_{i=1, \dots, n} (\Delta x_{i-1}) = \text{maximum width of collection of rectangles.}$

Definition of the Riemann integral.

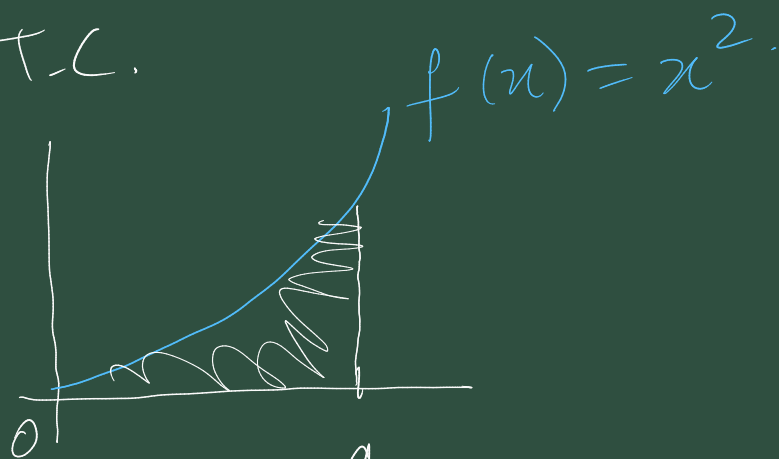
area of i^{th} rectangle.

$$\int_a^b f(x) dx := \lim_{\substack{n \rightarrow \infty \\ \Delta x \rightarrow 0}} \left(\sum_{i=1}^n \underbrace{f(x_i^*)}_{\text{height of } i^{\text{th}} \text{ rectangle}} \underbrace{\Delta x_{i-1}}_{\text{width of } i^{\text{th}} \text{ rectangle}} \right)$$

sum of areas of all rectangles.

This definition will both allow us to compute $\int_a^b f(x) dx$ for given functions f and also prove general properties of integration, such as the F.T.C.

Eg.



Calculate $\int_0^a x^2 dx$ using Riemann sum definition.

Choose a partition, the equally spaced partition.

• Notice: that $\frac{x^3}{3}$ is an anti-derivative of x^2 . which hints at a connection between indefinite and definite integrals, which will become clear with the F.T.C.

Inspect another example in APEX
eg. 5.3.24

$$\int_{-1}^5 x^3 dx = \dots = ~~48~~$$

$$= \lim_{n \rightarrow \infty} \left(156 + \frac{378}{n} + \frac{216}{n^2} \right)$$

$$= 156$$

Notice: again this agrees with expected result ~~from~~ using anti-derivative.

$$\left[\frac{x^4}{4} \right]_{-1}^5 = \frac{5^4}{4} - \frac{1}{4}$$

